

**CLASS-12**

**SENIOR SECONDARY EXAMINATION -2019**

**BIOLOGY**

**TIME: 3 ¼ HOURS**

**MM: 56**

**GENERAL INSTRUCTION TO THE EXAMINEES.**

- 1. Candidates must write first his/ her roll number on the question paper compulsorily.**
- 2. All questions are Compulsory.**

**Write the answer to each question in the given answer book only.**

**For questions having more than one part the answers to those parts are to be written together in continuity.**

**If there is any error difference contradiction in Hindi and English versions of the question paper the questions of Hindi version should be treated valid.**

**For questions having more than one part the answers to the other part should be written together in continuity.**

| 1. Section | q.no  | marks |
|------------|-------|-------|
| A          | 1-8   | 1     |
| B          | 9-14  | 2     |
| C          | 15-24 | 3     |
| D          | 25-30 | 5     |

- 2. There are internal services in question 27 to 30.**

**SECTION-A**

- 1. Define agamospermy.**

**[1]**

**Ans:** Agomosperry is the asexual reproduction of seeds and embryo in plants without fertilization or meiosis.

- 2. Deficiency of which element in the soil causes dieback disease of lemon (Citrus). [1]**

**Ans:** Deficiency of Copper element in the soil causes dieback disease of lemon (Citrus).

3. **Write the class of enzyme which catalyzes the oxidation and reduction reactions .[1]**

Ans: Oxidoreductases enzyme catalyzes the oxidation and reduction reactions.

4. **Write the name of plant hormone which is used to store potato tubers. [1]**

Ans: Plant hormone which is used to store potato tubers is Auxin.

5. **Write the name of technique to make lots of copies of DNA from a single copy in a very limited time.**

[1]

Ans: PCR is the technique to make lots of copies of DNA from a single copy in a very limited time.

6. **Define biogas. [1]**

Ans: Biogas is a type of biofuel that is naturally produced from the decomposition of organic waste. Generally dung is used for production of biogas in India.

7. **Which blood corpuscles are necessary for starting the process of blood clotting?**

[1]

Ans: Blood Platelets are necessary for starting the process of blood clotting.

8. **Name the point on retina where both rods and cones are absent. [ 1]**

Ans: At blind spot on retina where both rods and cones are absent.

9. **Write the name of hormone which prepare to face adverse condition in humans. [1]**

Ans: .

10. **What is the first cervical vertebra called? [1 ]**

Ans: AtlasVertebrae is the first cervical vertebra.

11. **Explain implantation. [1]**

Ans: After the process of fertilization the blastocyst sinks into a pit formed by Endometrium and then get completely buried in the endometrium that grows around it in the uterus .This step is called implantation.

12. **What is the measurement of haemoglobin present in blood called?**

[1]

Ans: Haemoglobinometry is used to measure haemoglobin present in blood.

**13. Write the full form of MRI.**

**[ 1]**

Ans: Magnetic Resonance Imaging.

### SECTION-B

**14. I) Write the relation between diffusion pressure deficit osmotic pressure and turgor pressure.**

Ans:  $DPT = OP - TP$

**ii) OP and DPT and equivalent in a flaccid Cell. Give reason. [1+1=2 ]**

Ans: There is no any pressure on cell wall in flaccid condition. Hence the cell has zero turgor pressure. At this time the Osmotic pressure of cytoplasm is equal to the diffusion pressure deficient(DPT).

**OR**

**i) Write the relation between water potential ( $\Psi_w$ ), osmotic potential( $\Psi_s$ ) and pressure potential  $\Psi_p$  .**

Ans:  $\Psi_w = \Psi_s + \Psi_p$

**ii) Calculate the water potential of solution which is enclosed in a semipermeable membrane whose osmotic potential is -35 bar and pressure potential is - 5 bar .**

Ans:  $-35 - 5 = -40$

**15. I) Name the critical elements present in the soil.**

Ans: Nitrogen, Phosphorus, potassium are the critical elements present in the soil.

**Ii) Deficiency symptoms of which element appears first in the apical bud.**

**[1+1=2]**

Ans: Deficiency symptoms of Calcium and Boron appears first in the apical bud.

**16. Lambda Bacteriophage is more important vector in comparison to M 13. Justify this statement giving two reasons.**

**[1+1=2]**

Ans: Lambda bacteriophage are more important vector than and M13 because

i) They are bacteriophages of E-coli

Ii) They possess linear and double helix DNA.

**17. Explain agrobacterium mediated gene transfer.****[2]**

Ans: This type of gene transfer is based on tumor inducing plasmid Ti plasmid. Ti plasmid is found in *Agrobacterium tumefaciens* found in soil and in dicots infected by this. During the infection of *A. tumefaciens* a copy of Ti plasmid gets integrated in the genome of host plant cell. The segment of DNA of Ti integrated in the genome host cell is called T-DNA or transfer DNA. In plants agrobacterium infection process is regulated by T1 plasmid and genes located of covalently closed circular DNA or CCC- DNA and chromosomes of agrobacterium. On both terminals of transfer DNA there is the sequence of direct repeats of 25 base pairs which is called the left and right terminal or the T-DNA. The gene transferred in the plant cell is introduced short indicated in the middle of the of these two terminals.

**18. Part A contain plant parts and part B contains useful plant parts related to them. Match the column A with Column B .** $\frac{1}{2} \times 4 = 2$ 

|           |                   |
|-----------|-------------------|
| Groundnut | Dry Flowerer Buds |
| Coconut   | Dried Rhizomes    |
| Cloves    | Endosperm         |
| Turmeric  | Seeds             |

Ans:

|           |                 |
|-----------|-----------------|
| Groundnut | Seeds           |
| Coconut   | Endosperm       |
| Cloves    | Dry flower buds |
| Turmeric  | Dried Rhizomes  |

**19. How blue green algae or cyanobacteria works as bio-fertilizer explains. ? [2]**

Ans: Cyanobacteria such as Anabaena, nostoc, Plectonema are prokaryotic non-symbiotic organism that take part in nitrogen fixation. They have specialized type of cells which are called as .Nitrogen fixation is done by and Nif genes present in heterocyst. The use of blue green algae increases the paddy yield so it is used as bio-fertilizer.

**20. Explain two respiratory disorders. [1+1=2]**

Ans: Two respiratory disorders are following –

- Asthma -This disease is caused due to allergy from pollen grains, dust particles from food stuffs, smoke, cold etc. In this disease cough and breathing becomes hard. in this disease mucus is formed in bronchi which causes swelling and narrowing and these develop hardness in breathing.
- Bronchitis- Due to the swelling on the inner surface of bronchus continued cough much mucus green yellow cough and hard breathing are its symptoms. This diseases caused by mainly by smoking.

**21. Write the name of hormone which induces the development of Corpus luteum .Explain the effect produced due to the damage of Corpus luteum.**

**[1+1=2]**

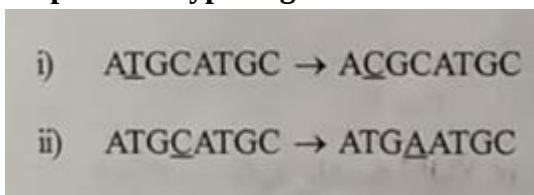
Ans: The Corpus luteum is maintained by the LH hormone secreted by the pituitary gland.

Progesterone hormone is released by Corpus luteum .The purpose of progesterone is to prepare the body for a pregnancy and maintain the pregnancy if it occurs .It results in development of the breast, increase in milk gland for physiological changes required for a pregnancy .if Corpus luteum is damaged there will be no pregnancy.

**22. Describe the structure of human sperm. [2]**

Ans: Sperm is a microscopic structure composed of head, neck and tail. A plasma membrane envelops the whole body of the sperm. The sperm head contains an elongated haploid nucleus the anterior portion of which is covered by a cap like structure called acrosome .the acrosome is filled with enzyme that help in fertilization of ovum. Head is followed by neck. The neck is followed by middle piece. It is composed of numerous mitochondria which produces energy that facilitates sperm motility. Next to middle piece is principal piece which ends in end piece.

23. Explain the type of gene mutation in each of the following. [1+1=2]



Ans:(i) It is transition type of gene mutation -

In this type of mutation replication of one Pyrimidine with another pyrimidine and one purine with another purine base takes place.

(II) Transversion-

In this type of mutation one purine base is replaced by another pyrimidine base or same type of pyrimidine base is replaced by some other purine base.

24. Name the pathogen of amoebic dysentery mention its two symptoms and two preventive measures.

[ 1+ ½+ ½ = 2]

Ans: Amoebic dysentery is caused by Protozoa name Entamoeba histolytica. in this disease trophozoite stage of pathogen enters the intestinal wall and secretes tissue decomposing enzymes which decompose the cell of tissues. This causes small bulges on these places in the form of small ulcer. Mucus gets filled in them. These come out with stool of faeces. Motion comes with cramps in stomach first and blood comes in motion. Patient becomes weak.

To get cure from disease water should be boiled properly before using it for drinking. Open food and other items should not be consumed. Houseflies' works as a mechanical carrier of this disease.

### SECTION - C

25. Describe the structure of and anatropous ovule with diagram.

[2+1=3]

Ans: In Anatropous ovule the funicle grows unilaterally and the ovule rotate 180° and becomes opposite in which the micropyle comes closer of the hilum and micropyle and chalaza lie in a straight line. 82% of families of angiosperms have this type of ovule.



**26. Define dihybrid cross. Explain Mendel dihybrid cross experiment by checker board. [1+1+1=3]**

Ans: When cross is done taking consideration the two pairs of factors or elements then it is called dihybrid cross. Mendel cross between pea plant having two pairs of contrasting characters. Yellow and round seed plants and green and wrinkled seeds plants. After crossing these plants all plants in the F<sub>1</sub> generation were developed yellow and round seeds in the pea plant. The yellow seed colour is dominant over green colour and round seed shape is dominant over wrinkled seed shape. When the seeds of F<sub>1</sub> generation were grown and the plants were self-fertilized then the following four types of seeds were obtained in F<sub>2</sub> generation whose phenotype ratio was

9 : 3 : 3 : 1

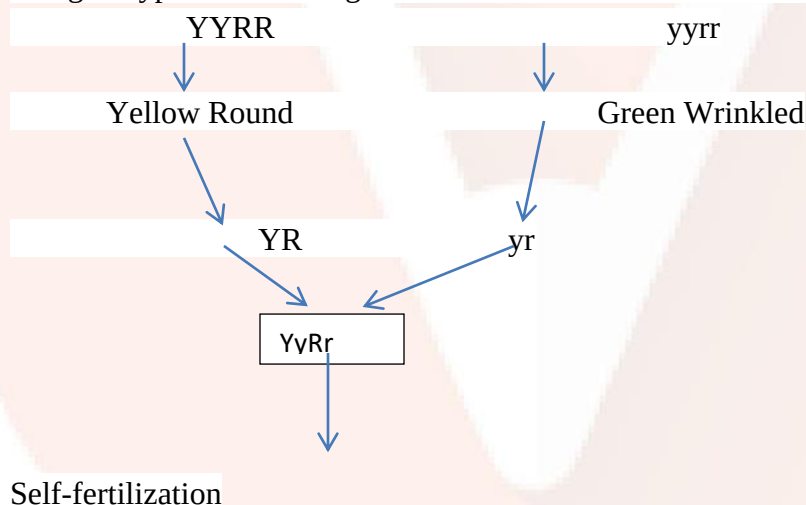
in 9/ 16 - both dominant traits -yellow and round seeds

in 3 /16 - one dominant and recessive trait - yellow and wrinkled seeds

3/16 – One recessive and one dominant trait- Green and Round seeds

In 1/16 – Both recessive traits- green and wrinkled seeds

The genotypic ratio of F<sub>2</sub> generation came = 1 : 2 : 2 : 4: 1 : 2 : 1 : 2 : 1



| ♀ \ ♂ | YR                                                                                        | Yr                                                                                        | yR                                                                                        | yr                                                                                        |
|-------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| YR    | <br>YYRR | <br>YYRr | <br>YyRR | <br>YyRr |
| Yr    | <br>YYRr | <br>YYrr | <br>YyRr | <br>Yyrr |
| yR    | <br>YyRR | <br>YyRr | <br>yyRR | <br>yyRr |
| yr    | <br>YyRr | <br>Yyrr | <br>yyRr | <br>yyrr |

### DIHYBRID Cross

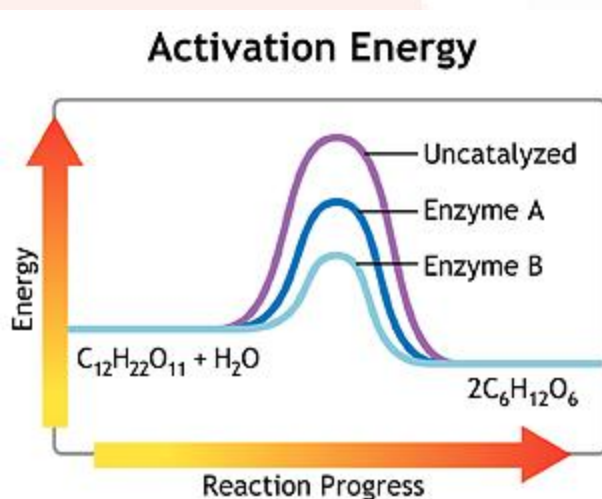
27. Explain the mode of enzyme action on the basis of lowering of activation energy.

Draw a labelled diagram to show the activation energy.

[2+1

=3]

Ans: In chemical reactions energy is released but not to initiate that reaction. Energy is also required for initiating the reaction these energy is called activation energy. Enzymes have this specialty that they can reduce the energy required for activation of substrate molecule. Thus in the presence of enzyme on lower activation energy substrate get converted into and products.

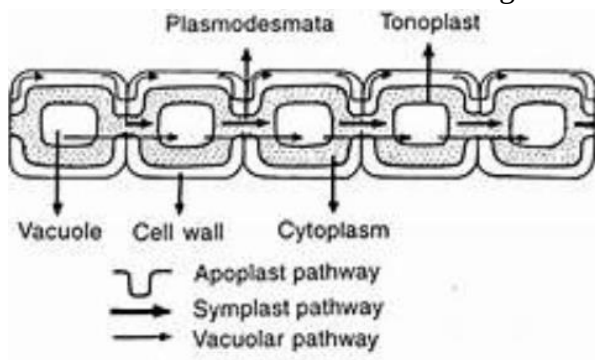


OR

Explain apoplast pathway of water in plants .draw a diagram to show measure parts of water flow in plant cells.



Ans: When the flow of water in plants occurs through nonliving cell wall and intercellular spaces present among cells then it is called apoplast pathway. This flow of water is uncontrolled and occurs through diffusion.



## SECTION -D

### 28. I) What is phosphorylation?

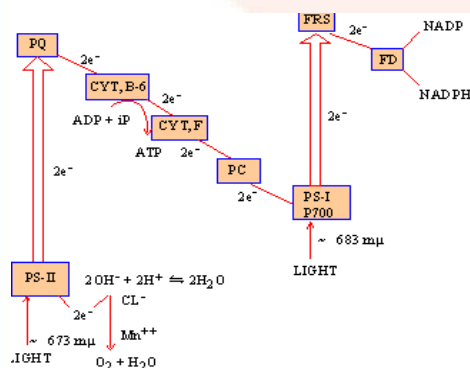
[1+2+1=4]

Ans: Chlorophyll molecules get excited for some time after absorption of energy packet or Quantum from electromagnetic waves. Hence electron released from chlorophyll molecule. Free electrons released from chlorophyll go through the different receptors. ATP synthesise from free energy due to this electron transfer is called photophosphorylation.

### ii) Explain cyclic photophosphorylation reaction.

Ans: In this reaction only PS1 (Photosystem1) takes place. Electron from  $P_{700}$  by light energy in PS1 passes through different electron acceptors A (FeS),  $Fd \rightarrow Cyt\ b_6 \rightarrow f$  through plastocyanin reaches back to  $P_{700}$ . Energy released between  $Cyt\ b_6 \rightarrow Cyt\ f$  used in the synthesis of ATP from ADP. In this way ATP is formed. In this reaction free electrons from  $P_{700}$  return back to  $P_{700}$  after synthesis of ATP. That is why it is known as cyclic photophosphorylation.

### iii) Draw a line diagram to show the reactions of cyclic photophosphorylation.



## CYCLIC PHOTOPHOSPHORYLATION

OR

I) Write the site of glycolysis in living.

Ans: The site of glycolysis is cytoplasm.

ii) Explain the various steps of phosphorylation of glucose in glycolysis.

**Ans:** In the first step of glycolysis one molecule of glucose in presence of hexokinase uses one molecule of ATP and form glucose 6 phosphate. This glucose-6-phosphate convert into fructose-6-phosphate in the presence of isomerase by isomerization. Fructose 6-phosphate again uses the one molecule of ATP and forms fructose 1,6-bisphosphate molecule in the presence of Phosphofructokinase.

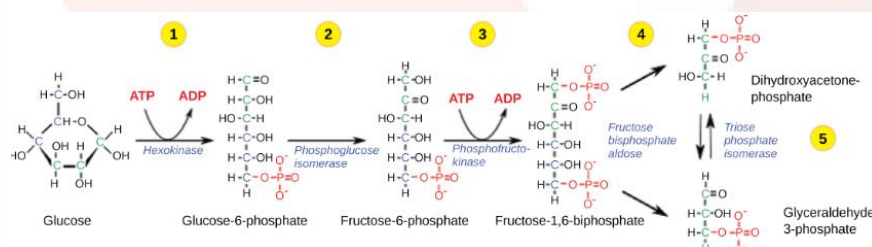
The reaction that takes place in the phosphorylation of glucose molecule are as follows:

glucose + ATP  $\xrightarrow{\text{Hexokinase, Mg}^{2+}}$  Glucose-6-phosphate + ADP(1)

Glucose-6-phosphate  $\xrightarrow{\text{Isomerase}}$  Fructose-6-phosphate

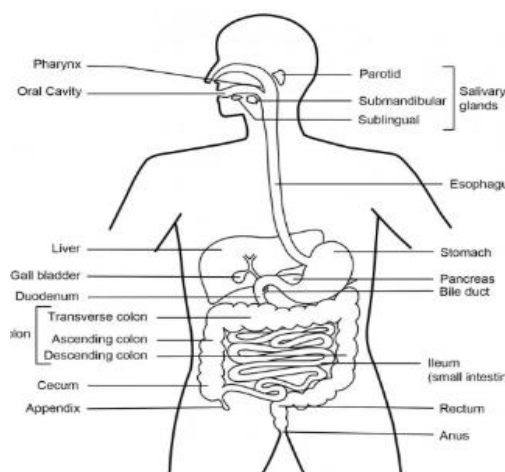
Fructose-6-phosphate + ATP  $\xrightarrow{\text{Phosphofructokinase, Mg}^{2+}}$  Fructose 1,6-bisphosphate

iii) Draw a line diagram of biochemical reactions upto phosphorylation of glucose.



29. Draw a labelled diagram of human digestive system. Explain the mechanism of digestion in stomach.

[1+3=4]

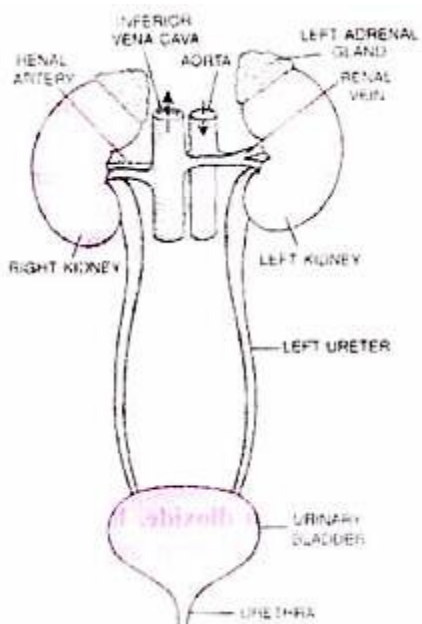


### HUMAN DIGESTIVE SYSTEM

The semi digested food direct in stomach for about 4 to 5 hours then gets mix with Acidic gastric juices by churning movement of a muscular wall and is called chyme at this stage chyme is a semi fluid mass into which food is converted by gastric secretion and which passes from the stomach into the small intestine. Gastric juice contains HCL, renin, mucous, pepsinogen and a little gastric lipase. Gastric juices are highly acidic. HCL lower down the pH of food .At this low pH activity of salivary lipase stops. Hydrochloric acid dissolve the bits of food and creates an acidic medium so that pepsinogen is converted into pepsin .Pessin is a protein digestive enzyme. It is inactive form called pepsinogen which then get activated by hydrochloric acid .the activated pepsin then converts protein into proteases and peptides. Renin is a proteolytic enzyme. In acidic medium it plays an important role in coagulation of milk.

**OR**

**Draw a labelled diagram of human excretory system.**



## EXCRETORY SYSTEM

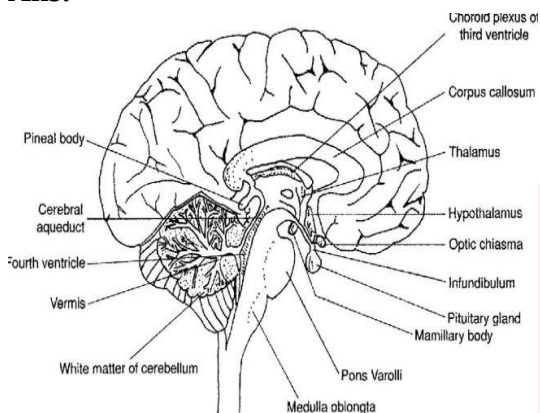
**Explain the mechanism of ultra filtration of urine formation.**

**Ans:** Bowman's capsule in the uniferous act function as a microseive. The afferent arteriole carries ureated blood in this and efferent arteriole takes away blood from this. The walls of capillaries of glomerulus has innumerable pores from which the permeability of porous membrane in comparison to blood capris increases .The substance dissolved in plasma can get through this pores .The afferent arteriole entering the glomerulus is more in diameter that efferent arterioles. So blood flows fast in this, but at the time going out would not flow so fast. In a healthy person the pressure of blood coming in the glomerulus is more. Blood corpuscles blood proteins and blood soluble colloids could not filter .An opposite blood colloidal osmotic pressure is produced by Colloidal substances. In the blood the filtrate in Bowman's capsule cavity also produces about 20 mm Hg pressure which is called capsular hydrostatic pressure .Thus the net effective filtrate pressure on glomerulus fluid is about 20 mm Hg which works and the fluid plasma excretory substances remain moving in the cavity of Bowman's capsule. The filtered plasma from the glomerulus is called glomerular filtrate. This fluid comes down into the cavity of bowman's capsule .the process of filtration from the wall of capillaries of glomerulus of excretory and other utilizable substances is known as ultrafiltration.

30. I) Draw a labelled diagram of human brain.

[1+2+1=4]

Ans:



ii) **Describe cerebrum.**

**Ans:** Cerebrum is the most developed and biggest faster part of the forebrain. This is divided into two parts left and right which are called cerebral hemisphere. These both hemispheres are connected with each other with heterogeneous nerve ray sheets called Corpus callosum. On its outer surface due to presence of ridges and grooves it is not smooth. Cerebrum appears oval shaped when it is seen from the top it is a Fissure in the middle which separates both the hemisphere. Each hemisphere is empty inside and its wall has two region- external and internal. External region cerebral hemisphere cortex has neuron of Nerve cells and as it is grey therefore known as grey matter. Internal region consists of Axon and it is called white matter.

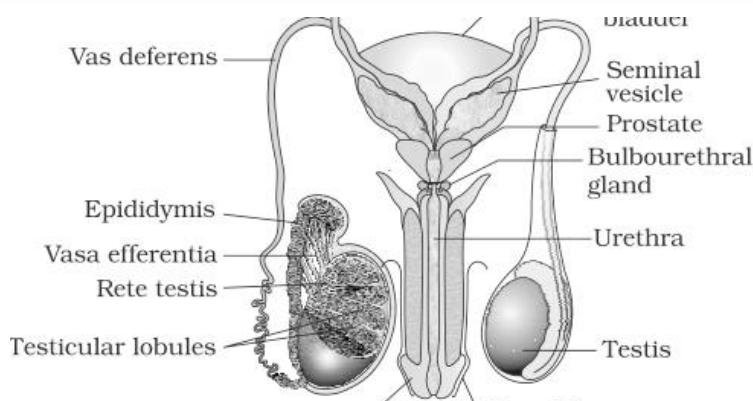
iii) **Write two functions of cerebrum.**

Functions of cerebrum-

It performs the function of planning, memorizing, thinking, argument and comment. It receives and analyses the information from sensory organs like nose and ear and perform the function accordingly.

**OR**

i) **Draw a labeled diagram of Male reproductive system.**



## MALE REPRODUCTIVE SYSTEM

### ii) Describe the structure of testes.

Ans: In an adult man there are a pair of pink and oval shaped testis which are situated outside the abdominal cavity and in the middle of leg. Each testis is about 4cm thick and 2.5 cm long structure having 10- 15 grams of weight. Testis has two covers which are called testicular capsules. Each testis is divided into 250 to 300 lobules. In each lobule there are two or three thin and coiled seminiferous tubule are present. In the lobular tissue besides this tubules endocrine cells blood capillaries and nerve fibres are also present besides this interstitial cells or Leydig cells are present in connective tissue which secrete male hormone testosterone due to which secondary sexual characters and accessory sex gland develop in males.

### iii) Write two functions of Sertoli cell.

Ans: Sertoli cells provide support and nourishing disintegrate the unnecessary cytoplasm of spermatids. These also secrete the protein hormone inhibin which regulates the function of follicle stimulating hormone.